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Joint Antenna Selection and Proportional Fairness User Scheduling for Multi-User Massive MIMO Systems

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Abstract: Massive multi-input multi-output (massive MIMO) technology offers significant multiplexing gains and enhances transmission rates by efficiently utilizing available airspace resources. However, it requires each antenna to be paired with a separate radio frequency (RF) chain, which leads to the need for numerous RF chains in the system, resulting in high hardware costs, increased computational complexity, and elevated power consumption. To address this, antenna selection technology reduces the number of RF chains required, activating only the antennas that correspond to the available RF chains. Moreover, user scheduling provides multi-user diversity in multi-user massive MIMO systems. Therefore, this paper introduces a joint antenna selection and orthogonality-based user scheduling (JAS-OUS) algorithm aimed at maximizing the system sum rate. Furthermore, to tackle the issue of fairness, which is often overlooked by traditional user scheduling algorithms, a proportional fairness user scheduling (PFUS) approach is proposed. In this scheme, user weights are updated based on proportional fairness, ensuring a fair selection of users for communication in each time slot. Simulation results demonstrate that the JAS-OUS algorithm achieves robust performance across various configurations of transmitting antennas and users. Additionally, when combined with PFUS, the joint algorithm ensures more equitable user participation in communication without compromising the system sum rate.

Keywords: massive MIMO; antenna selection; user allocation; proportional fairness

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Academic Editor: Yutaka Ishibashi

Received: 19 February 2025
Revised: 18 April 2025
Accepted: 23 April 2025
Published: 28 April 2025

Citation: Chen, B.; Yang, L.; Wu, M. Joint Antenna Selection and Proportional Fairness User Scheduling for Multi-User Massive MIMO Systems. *Appl. Sci.* **2025**, *15*, 4916. https://doi.org/10.3390/app15094916

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1. Introduction

Massive MIMO technology leverages a large number of antennas to improve communication performance. By deploying numerous antennas at both the base station and user sides, this technology enhances signal robustness against interference, increases transmission rates, and improves user connectivity [1–3]. Due to its massive connectivity, high energy efficiency, and low latency, massive MIMO has become one of the key technologies for building networks such as the Internet of Things (IoT) [4,5].

However, in a massive MIMO system, each transmission antenna is paired with an individual RF chain, resulting in the number of RF chains being equal to the number of antennas. This setup leads to increased computational complexity and power consumption. A major solution is to perform transmission antenna selection. In this architecture, only a limited number of RF chains are deployed, and a corresponding number of antennas

are activated for communication based on the available RF chains [6,7]. By adopting this approach, both power consumption and latency are significantly reduced. Notably, low power consumption and low latency are critical performance metrics for 5G and beyond 5G (B5G) networks [8]. Reference [9] proposed a greedy algorithm for fast selection of transmit or receive antennas, which significantly reduces computational complexity without compromising capacity performance. In the context of massive MIMO, Reference [10] jointly optimizes precoding and antenna selection. A low-complexity cyclic binary meta-heuristic algorithm is employed to select the optimal antenna combination, thereby enhancing energy efficiency while maintaining spectral efficiency.

In addition, user scheduling plays a crucial role in multi-user large-scale MIMO systems, as it enhances multi-user diversity gain and improves communication efficiency [11]. Reference [12] investigates multi-user scheduling in 6G vehicle-to-everything (V2X) communications. Block diagonalization precoding is employed to eliminate multi-user interference, and a group of users with low correlation and high channel quality is selected for scheduling to enhance the overall channel gain and reduce the bit error rate. Reference [13] addresses the scheduling problem in multi-user MIMO broadcast channels, where user scheduling and antenna selection are performed under limited feedback conditions to reduce channel feedback overhead.

Furthermore, the combination of antenna selection and user scheduling enables the optimal pairing of antennas and users, resulting in a more favorable channel matrix and improved data transmission rates under the same spectrum resources. User scheduling can reduce inter-user signal interference and optimize the system's signal-to-noise ratio (SNR). In massive MIMO systems, antenna selection significantly reduces the signal processing burden and hardware cost, contributing to the realization of green communication.

The advantages of antenna selection and user scheduling have led to their widespread application in practical communication scenarios. In 5G/6G massive MIMO base stations, antenna selection is used to reduce the number of RF chains, while user scheduling enables multi-user parallel transmission, thereby improving system throughput. In the IoT, user scheduling controls active nodes to reduce concurrent interference. In V2X and high-mobility scenarios, user scheduling dynamically adjusts the service order based on vehicle speed, location, and service priority, reducing latency and ensuring reliable communication (e.g., for autonomous driving). In unmanned aerial vehicle (UAV) communications, antenna selection enables directional transmission, while user scheduling is carried out based on the ground user density.

In the design of joint antenna selection and user scheduling, the conventional approach is to separate these two processes and implement them in a staged manner. While this strategy reduces computational complexity, it comes at the cost of degraded system sum rate [14]. In contrast, joint optimization can more effectively exploit spatial resources by selecting the optimal subsets of users and antennas under limited resources (e.g., RF chains, transmit power), thereby maximizing the system's sum rate.

Successive elimination is a commonly used antenna selection method [15–18]. It starts by assuming that all antennas are involved in communication and then eliminates one antenna with the least contribution to the system in each iteration until the desired number of antennas is achieved. Benmimoune et al. propose a joint strategy of base station antenna selection and user allocation in large-scale MIMO broadcast channel, aiming to optimize the total system throughput [15]. By successively eliminating the antennas with the least contribution to the system and using semi-orthogonal user selection (SUS) algorithm to obtain a set of optimal users simultaneously, a sum rate performance close to the exhaustive search method can be achieved with very low complexity. The user scheduling algorithm is changed in reference [16]. First, users whose channel singular values are greater than

the preset threshold are selected to ensure communication efficiency. Then, a detailed user scheduling is carried out in this user group. The user with the largest singular value is the first scheduled user, and the remaining scheduling users are selected according to the principle of low orthogonality with all the selected users. In reference [17], a low-complexity suboptimal user allocation approach assisted by a matrix Gaussian elimination method (MGEM) is proposed. The user scheduling scheme is integrated with norm-maximized antenna selection to optimize the sum rate, subject to the constraints of a maximum number of served users and limited transmission power. In reference [18], a set of transmitting antennas and a group of users which can generate maximum channel matrix eigenvalues are selected at first. Then, the antennas and users are iteratively updated. The worst antenna or user is eliminated at each time and replaced by the antenna or user that contributes most to the sum rate. Finally, a maximum system sum rate can be achieved. The difference in this scheme lies in selecting a suboptimal user group and antenna group in the beginning. The user group and antenna group are constantly optimized to approach the optimal scheme. In reference [19], the antennas are pre-divided into multiple antenna groups, with the antennas within each group working cooperatively. A greedy strategy is employed to select the optimal group from the candidate antenna groups, while simultaneously scheduling the user group that best matches the selected antenna group. Compared to the traditional per-antenna selection approach, this method significantly reduces the search space.

The primary focus of the above studies is to achieve favorable system sum rate with low computational complexity. In addition, there are studies on different channel state information as well as system models. When only part of channel state information (CSI) is known, a joint antenna selection and user allocation algorithm on the base station side using the partially observable Markov decision processes (POMDP) framework is designed in reference [20]. At the start of every frame, the transmitter schedules the users participating in communication and selects the corresponding antenna subset to serve each group of users. For massive MIMO downlink time-varying fading channels, Zhu et al. study the problem of weighted average sum rate maximization, while guaranteeing the lowest average sum rate for users [21]. Adaptive scheduling of antennas and users is implemented through dual and randomized subgradient approach, with a heuristic algorithm applied in each time slot to minimize computational complexity. In a multi-cell large-scale MIMO scenario, a resource-efficient joint antenna and user selection technique leveraging the Adaptive Markov Chain Monte Carlo (AMCMC) algorithm is reported in reference [22]. The algorithm considers multi-cell interference and selects antennas and user subsets to optimize the cell capacity. In reference [23], the joint optimization problem in non-orthogonal multiple access (NOMA) systems is investigated. For a multi-band, multi-user scenario, a joint antenna and user contribution algorithm is proposed, which jointly considers the contribution of each antenna's and each user's channel gain to the overall channel gain. The proposed method is shown to be more effective in scenarios with high user density and a large number of antennas. All the above studies involve antenna selection on base station side. Sigdel et al. investigate antenna selection on user side in [24]. A user grouping strategy utilizing subspace similarity is proposed, with the goal of selecting users with better spatial compatibility to improve the system throughput. Furthermore, to address the issue of zero rates for certain users caused by user scheduling, a proportional fairness (PF) scheduling scheme is introduced.

Table 1 lists the aforementioned related works and summarizes their system models, research objectives, and methods.

Table 1. Comparison of related works.

Reference	System Model	Research Objective	Method
[15–18]	Multi-user Massive MIMO, Single-antenna user	Maximize the system sum rate under low computational complexity	Greedy antenna elimination, SUS
			Greedy antenna elimination, User preselection based on singular values, SUS
			MGEM-based user selection, maximum norm antenna selection
			User scheduling based on the squared norm of the channel matrix
[19]	Massive MIMO, antenna group, CSI feedback	Reduce feedback overhead while ensuring system sum rate	Antenna group selection is performed first, followed by intra-group user scheduling
[20,21]	Massive MIMO, time-varying channel	Maximize the long-term average sum rate (long-term performance)	POMDP-based joint design
		Maximize the system weighted average sum rate while ensuring user QoS requirements	OJAUS and low-complexity heuristic JAUS
[22]	Multi-cell, massive MIMO	Maximize the total sum rate of each cell under multi-cell interference constraints	AMCMC
[23]	Massive MIMO-NOMA, multi-user multi-band	Efficiently allocate antennas and users across multiple frequency bands to maximize the sum rate	Joint antenna and user contribution algorithm, considering user pairing and power allocation in NOMA
[24]	Multi-antenna users, user-side antenna selection	Maximize the weighted sum rate	PFUS under block diagonalization and successive optimization precoding schemes

From the above analysis, it can be seen that the joint antenna selection and user scheduling scheme suffers from the fact that the transmitter always tries to prioritize the allocation of resources to some users with the best channel conditions, thus neglecting the service quality for the remaining participants. This kind of service lacks fairness regarding the system, because some users with poor channel conditions may never be selected to participate in communication.

Compared with traditional maximum-rate scheduling, which prioritizes users with high channel quality, PF scheduling exhibits a notable ability to balance system performance and user fairness. This scheduling strategy dynamically adapts to time-varying channel conditions by jointly considering both the current channel state and the historical resource allocation of users. In each scheduling instance, it compensates users who have not been served for an extended period. By doing so, PF scheduling achieves a trade-off optimal solution that ensures both a high system sum rate and fair service distribution among users. Moreover, PF scheduling is inherently compatible with complex network architectures such as orthogonal frequency division multiple access (OFDMA), multi-user MIMO, and 5G NR. Therefore, after investigating the low-complexity joint antenna selection and user scheduling algorithm JAS-OUS, we propose a transmission scheme named JAS-PFUS, which integrates PF scheduling to ensure fairness in user scheduling.

The key contributions of this paper are outlined below.

1. A system framework is constructed, where a base station fitted with multiple antennas communicates with several users equipped with one antenna. In order to realize simultaneous antenna and user selection strategy, an optimization task is defined with the goal of maximizing the system throughput. N_{RF} antennas are selected from N_t transmitting antennas, and K users are selected from U participants at the same instant.
2. We propose a joint antenna selection and channel orthogonality-based user scheduling algorithm, JAS-OUS, which leverages the spatial selectivity benefit from antenna allocation and multi-user diversity benefit from user allocation correspondingly, to optimize the system sum rate. The key idea of the introduced algorithm is to sequentially remove the weakest antenna, that is, the antenna with the least contribution to the system throughput. Also, the user set having the smallest channel correlation is selected using orthogonality-based user scheduling (OUS) for each antenna removal.
3. Further, in order to ensure the fairness of user scheduling, an improved PFUS user allocation strategy is introduced with the goal of maximizing the weighted throughout. An adjustable weight is set for each user so that different users are selected for communication in each time slot, which makes sure more users will have the opportunity to be served.
4. Simulation results and complexity assessments indicate that JAS-OUS has the potential to guarantee a high transmission rate with lower computational complexity. Compared with the traditional non-proportional fairness scheduling algorithm, the proposed JAS-PFUS algorithm shows good fairness and is able to schedule more users in communication without reducing the system sum rate.

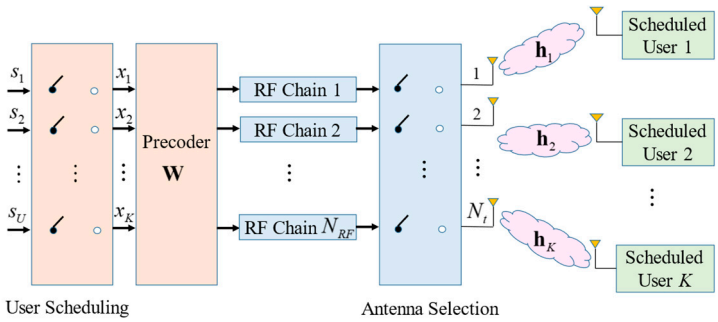
The paper is structured as follows. In Section 2, we introduce the joint antenna selection and user allocation model for multi-user massive MIMO and derive the system throughput expression. The specific JAS-OUS and improved JAS-PFUS algorithms are described in Section 3. In Section 4, the simulation data and complexity analysis are given. Section 5 provides the conclusion of this paper.

We use the following notation throughout this paper: $\mathbf{A}$ is a matrix; $\mathbf{a}$ is a vector; a is a scalar; $\mathcal{A}$ is a set. $(\cdot)^T$ and $(\cdot)^H$ denote transpose and conjugate transpose, respectively. $|\mathcal{A}|$ denotes the cardinality of set $\mathcal{A}$ and $\|\mathbf{a}\|_2$ denotes the Euclidean norm of vector $\mathbf{a}$. $\mathcal{A}\backslash\{i\}$ represents the set $\mathcal{A}$ with element i removed. $\mathbf{H}(i,:)$ represents the ith row of matrix $\mathbf{H}$. $\mathbf{H}_{ij}$ denotes the element in the ith row and jth column of the channel matrix $\mathbf{H}$.

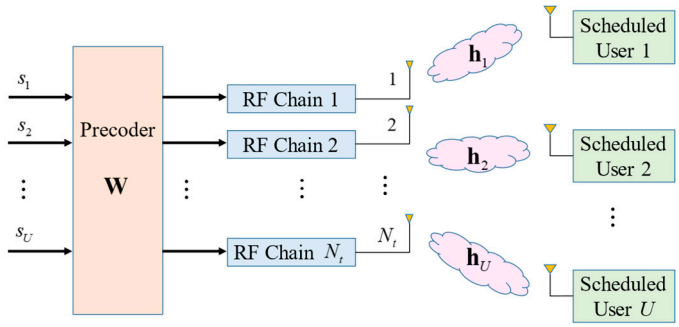
2. System Model

This paper considers a multi-user massive MIMO communication system, as illustrated in Figure 1a. The base station is equipped with N_t transmit antennas, and there are U single-antenna users. To reduce energy consumption and hardware cost while improving the system rate in a multi-user scenario, antenna selection and user scheduling are employed. The original transmitted signal consists of U user information signals, denoted by $\mathbf{s} \in \mathbb{C}^{U\times 1}$. User selection is realized through a switch array, which disables part of the data streams and retains only K streams for transmission, forming the signal $\mathbf{x} \in \mathbb{C}^{K\times 1}$. This selected signal is then digitally precoded to eliminate inter-user interference. The precoded signal is converted from digital to analog by N_{RF} RF chains, where $K \leq N_{RF}$. Finally, the signal is delivered to the transmit antennas through an RF switch array, which deactivates part of the antennas and only retains N_{RF} active antennas for communication. In this way, antenna selection and user scheduling are jointly accomplished. We define the group of selected transmitting antennas as $\mathcal{A}$ and the group of scheduled users as $\mathcal{S}$. The use of switch-based hardware in the system design is due to its compact size, high integration,

and reliability, which make it a practical solution for selectively disabling data streams and deactivating antennas.



(**a**)



(**b**)

Figure 1. Multi-user massive MIMO system model: (**a**) antenna selection and user scheduling system; (**b**) traditional fully digital system.

Figure 1b illustrates a conventional fully digital system. In this system, neither user scheduling nor antenna selection is performed. The absence of user scheduling prevents the exploitation of multi-user diversity gain, resulting in a degradation of the average system throughput. Moreover, if the users' channels are highly correlated, transmitting signals to multiple users simultaneously will lead to severe inter-user interference. The absence of antenna selection implies that each antenna in the system is connected to a dedicated RF chain, i.e., $N_{RF} = N_t$, resulting in an extremely high hardware cost and power consumption. In contrast, in the system shown in Figure 1a, only the selected antennas are equipped with complete RF chains (e.g., digital-to-analog converters/analog-to-digital converters, amplifiers), while the remaining antennas can be ignored, i.e., $N_{RF} \ll N_t$. This reduces the number of RF chains, thereby lowering hardware cost, reducing power consumption, and simplifying signal processing complexity. Furthermore, when certain antennas suffer from hardware faults or poor channel conditions, the system can still maintain robust performance by selecting antennas in better states, thereby improving overall system robustness.

In the antenna selection and user scheduling system, $\mathbf{x} = [x_1, x_2, \ldots, x_K]$ denotes the data streams sent to K scheduled users. After user scheduling and antenna selection, the received signal for K users could be represented as

$$\mathbf{y} = \mathbf{HWx} + \mathbf{n}, \tag{1}$$

$\mathbf{H} = \left[\mathbf{h}_1^T, \mathbf{h}_2^T, \ldots, \mathbf{h}_K^T\right]^T$ is the channel matrix formed by all scheduled users and the selected antennas, where $\mathbf{h}_i \in \mathbb{C}^{1 \times N_{RF}}$ stands for the channel state response between the

base station and the scheduled user i. $\mathbf{h}_i$ is considered to be an independent and identically distributed Rayleigh fading channel. To reduce the interference among different users, the forced-zero precoding is used [25]. $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k]$ is the precoding matrix and $\mathbf{W} = \mathbf{H}\left(\mathbf{H}^H\mathbf{H}\right)^{-1}$. $\mathbf{n}$ denotes the noise vector, $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_K)$.

Since zero-forcing precoding is employed, the inter-user interference is eliminated. Then, the signal obtained by the scheduled user i can be written as

$$y_i = \mathbf{h}_i\mathbf{w}_i x_i + n_i,\ i = 1, 2, \ldots, K. \tag{2}$$

The SNR at the scheduled user i is

$$\gamma_i(\mathcal{A}, \mathcal{S}) = \frac{p_i \mid \mathbf{h}_i\mathbf{w}_i \mid^2}{\sigma_i^2}, \tag{3}$$

where p_i denotes the user's allocated power and σ_i^2 is the noise power.

The highest achievable rate for the scheduled user i is

$$R_i(\mathcal{A}, \mathcal{S}) = \log_2(1 + \gamma_i(\mathcal{A}, \mathcal{S})). \tag{4}$$

Further the total throughput expression can be calculated as [25]

$$R_{sum}(\mathcal{A}, \mathcal{S}) = \sum_{i=1}^{K} R_i(\mathcal{A}, \mathcal{S}) = \sum_{i=1}^{K} \log_2\left(1 + \frac{p_i \mid \mathbf{h}_i\mathbf{w}_i \mid^2}{\sigma_i^2}\right). \tag{5}$$

3. Methods

3.1. Problem Formulation

In order to optimize the joint antenna selection and user allocation, the maximization problem is formulated, which aims to select N_{RF} antennas from N_t transmitting antennas and select K users from all U users to obtain a maximum sum rate.

The joint antenna selection and user allocation problem is formulated as

$$\begin{aligned} &\max_{\mathcal{A},\mathcal{T}} R_{sum}(\mathcal{A}, \quad \mathcal{S}) = \max_{\mathcal{A},\mathcal{S}} \sum_{i=1}^{K} \log_2(1 + \gamma_i(\mathcal{A}, \mathcal{S})) \\ &\text{subject to} \qquad |\mathcal{A}| = N_{RF}, \\ &\qquad\qquad\quad |\mathcal{S}| = K, \\ &\qquad\qquad\quad \sum_{i=1}^{K} p_i \leqslant P_{total}, \end{aligned} \tag{6}$$

where P_{total} denotes the overall transmission power. p_i is determined by water filling method.

The optimization objective of the water-filling method is to determine the power allocated to each scheduled user in order to achieve a considerable system sum rate [26].

3.2. Joint Antenna Selection and Orthogonality-Based User Scheduling

Within this subsection, the JAS-OUS algorithm is introduced, which takes advantage of the spatial selectivity and multi-user diversity, to achieve maximum system sum rate. The basic idea of the introduced scheme is to remove the antenna contributing the least to the system in each iteration. At the same time, the users who exhibit good orthogonality in each iteration are selected to form a subset of users. The algorithm is terminated when the final number of selected antennas is N_{RF}.

The algorithm takes the channel matrix $\mathbf{H}$ as input and adopts the classical successive elimination method for antenna selection. It begins with initialization, assuming that all transmit antennas are initially selected, i.e., $\mathcal{A} = \{1, 2, \ldots, N_t\}$. Then, the algorithm per-

forms $N_t - N_{RF}$ iterations, where in each iteration, one antenna with the least contribution is removed. Eventually, only N_{RF} antennas are retained for signal transmission. In each iteration, all currently active antennas are examined. For each antenna, it is temporarily deactivated, and the OUS algorithm is used to determine the best user group under the current antenna configuration. Based on the selected antenna–user combination, the system sum rate is computed. By comparing the sum rates across all possible antenna deactivations, the antenna whose removal results in the highest sum rate is identified as having the least contribution to the system and is thus removed. After $N_t - N_{RF}$ iterations, the final set of selected antennas $\mathcal{A}$ and the set of scheduled users $\mathcal{S}$ are obtained. The pseudocode of JAS-OUS is given by Algorithm 1.

Algorithm 1. JAS-OUS

Input:
Channel matrix $\mathbf{H}$;
Number of RF chains N_{RF};
Transmit antenna N_t;
1: Initialization:
$t = 1$;
$\mathcal{A} = \{1, 2, \ldots, N_t\}$;
2: while $t < (N_t - N_{RF})$ do
$maxRate = 0$;
3: for each i in $\mathcal{A}$, do
$\tilde{\mathbf{H}} \leftarrow \mathcal{A} \backslash \{i\}$;
$\mathcal{S} \leftarrow$ choose K users using OUS $\left(\tilde{\mathbf{H}}, N_{RF}\right)$;
$R_{-i} = R_{sum}(\mathcal{A} \backslash \{i\}, \mathcal{S})$;
4: if $R_{-i} > maxRate$, then
$maxRate = R_{-i}$;
$i_{worst} = i$;
end
end
5: $\mathcal{A} = \mathcal{A} \backslash \{i_{worst}\}$;
$t = t + 1$;
end
Output:
the set of selected antennas $\mathcal{A}$;
the set of scheduled users $\mathcal{S}$

The user scheduling part is given by the orthogonality-based user scheduling algorithm. It takes channel $\tilde{\mathbf{H}}$ as input, which is formed by removing the current ith transmitting antenna, and finally outputs the group of scheduled users $\mathcal{S}$, which contains K users. First, initialization is performed so that $\mathcal{S}$ is an empty set. The first user is scheduled according to a rule that selects the one with the maximum channel power. Next, scheduling is performed for the rest of the users. At each iteration, a user is selected to be included in set $\mathcal{S}$. The selection criterion is to minimize the orthogonality coefficient between the current round's selected user and all previously chosen users, i.e., to ensure good orthogonality between this selected user and the previously selected users. The operation is repeated until K users are selected to form the final set of scheduled users. Algorithm 2 provides the pseudocode of OUS.

Algorithm 2. OUS

Input:

Channel matrix $\tilde{\mathbf{H}}$;

Number of users to be selected K;

1: Initialization:

$X = \{1, 2, \ldots, U\}$;

$\mathcal{S} = \varnothing$;

$n = 1$;

2: $I = \underset{i \in X}{\operatorname{argmax}} norm\left(\tilde{\mathbf{H}}(i,:)\right)$;

$s_1 = X(I)$;

$\mathcal{S} = \mathcal{S} \cup \{s_1\}$;

$X(I) = \varnothing$;

$n = n + 1$;

3: while $2 \leqslant n \leqslant K$ do

$M = \text{numel}(X)$;

4: while $1 \leqslant j \leqslant M$ do

$$b(j) = \sum_{c=1}^{n-1} \frac{|\tilde{\mathbf{H}}(X(j),:) * \tilde{\mathbf{H}}(\mathcal{T}(c),:)|^2}{|\tilde{\mathbf{H}}(X(j),:)|^2 * |\tilde{\mathbf{H}}(\mathcal{T}(c),:)|^2};$$

end

$I = \underset{j}{\operatorname{argmin}} b(j)$;

5: $s_n = X(I)$;

$\mathcal{S} = \mathcal{S} \cup \{s_n\}$;

$X(I) = \varnothing$;

$n = n + 1$;

end

Output:

the set of scheduled users $\mathcal{S}$

3.3. Proportional Fairness Based Joint Antenna Selection and User Scheduling

The user scheduling scheme aiming at sum rate maximization has the shortcoming that the base station always tries to prioritize the allocation of resources to some users with the best channel conditions, thus ignoring the service levels for the remaining users. This kind of service is unfair to the whole system, since some users with poor channel conditions will never be selected to participate in communication. Proportional fair scheduling, on the other hand, schedules different users in each time slot according to user weights, and more users can be guaranteed to participate in the communication during the whole communication process, thus achieving the balance between communication efficiency and fairness. Therefore, we introduce a joint transmitting antenna selection and proportional fair user allocation scheme to schedule more users during the whole communication process.

In user scheduling for communication systems, to ensure user fairness while improving system capacity, several fairness-oriented scheduling strategies are commonly employed in addition to PF scheduling. These include round-robin scheduling, max-min fairness scheduling, and weighted fair queuing, among others. Round-robin scheduling allocates the channel to users in a fixed cyclic order, regardless of their current channel conditions. It is simple to implement and offers good user fairness, but tends to ignore channel quality, resulting in lower system throughput. Max-min fairness scheduling prioritizes users with the lowest current transmission rate, maximizing their resource allocation. This approach is especially favorable to low-rate users and is suitable for scenarios with

strict quality of service (QoS) requirements, though it may sacrifice overall system throughput. Weighted fair queuing allocates resources based on assigned weights reflecting user priorities or data rates, offering a flexible mechanism to meet differentiated QoS demands. Among these methods, the adopted PF scheduling achieves a balance between fairness and efficiency, ensuring acceptable fairness across users while still maintaining a high system sum rate.

In order to fully reflect the differences between users, we first define a weight for each user, indicating the importance of the user. Assuming the rate of user i during the current time slot t is $r_i(t)$ and the average rate for the whole period is $\bar{r}_i(t)$. The weight $\lambda_i(t)$ for user i is expressed as

$$\lambda_i(t) = \frac{r_i(t)}{\bar{r}_i(t)}. \tag{7}$$

The average rate of every single user must be renewed to adjust the user weights, and the update rule is shown as follows [27],

$$\bar{r}_i(t) = \begin{cases} \left(1 - \frac{1}{T_c}\right)\bar{r}_i(t-1) + \frac{1}{T_c} r_i(t-1), & i = k, \\ \left(1 - \frac{1}{T_c}\right)\bar{r}_i(t-1), & i \neq k, \end{cases} \tag{8}$$

T_c denotes the update time window parameter or is understood as the number of time slots. It represents the average rate is calculated over a period of time. The update time window parameter is determined by actual scenario. When T_c is large, the calculation of a user's average rate needs to take into account the rate value of a long period of time in the past, which is more likely to reflect the long-term fairness. When T_c is small, it indicates that the calculation of a user's average rate only takes into account the rate value of more recent time slots, which is not universal.

In addition, when a user is always scheduled to participate in communication, its average rate will keep increasing. According to the expression of user weight, with the increase in average rate, the user weight will decrease, which means that the user is less important. On the other hand, if a user is not always scheduled, its average rate will keep decreasing and the corresponding weight will increase, which improves the importance of the user and increases the possibility that he is scheduled. Therefore, by adaptively adjusting the user weight, more users can be scheduled to participate in communication.

Due to the introduction of user weight parameters, the final objective function of joint antenna selection and user allocation needs to be adjusted accordingly. By adapting the optimization problem to weighted sum rate maximization, we have

$$\max_{\mathcal{A},\mathcal{T}} \hat{R}_{sum}(\mathcal{A}, \mathcal{S}, t) = \max_{\mathcal{A},\mathcal{T}} \sum_{i=1}^{K} \lambda_i(t) r_i(t), \tag{9}$$

where $\lambda_i(t)$ denotes the weight of the ith user in the current time slot t. $r_i(t)$ denotes the rate in time slot t. $\mathcal{A}$ and $\mathcal{S}$ are the group of selected antennas and the group of scheduled users, respectively.

The pseudocode of the joint antenna selection and proportional fairness user allocation is presented in Algorithm 3. All the user weights are initialized to 1. The Algorithm schedules users first. It utilizes the water filling method to calculate the unweighted rate $r_i(t)$ of all users in the current time slot t with all transmitting antennas participating in communication. The weighted rate of the ith user is given by $\lambda_i(t) r_i(t)$. Then, K users with the largest weighted rate are scheduled to constitute the set $\mathcal{S}$ during the current time slot. Fixing the set of scheduled users $\mathcal{S}$ for time slot t, the algorithm utilizes successive eliminating method as Algorithm 1 for antenna selection. The antenna has the smallest

contribution to the system throughput is eliminated in each iteration until reaching the specified number of antennas, with a total of $N_t - N_{RF}$ iterations. Once user allocation and antenna selection have been completed, the weights $\lambda_i(t)$ are updated for next time slot. Lastly, the whole process above is repeated for T_c times because user allocation and antenna selection are carried out in each time slot.

Algorithm 3. JAS-PFUS

Input:
Channel matrix $\mathbf{H}$;
Number of users to be selected K;
Number of RF chains N_{RF};
Transmit antenna N_t;
1: Initialization:
$X \leftarrow \{1, 2, \ldots, U\}$;
$\mathcal{S} \leftarrow \varnothing$;
$T_c = 20$;
$t = 1$;
2: User scheduling:
obtain rate per user $r_i(t)$ by water filling;
$\mathcal{S} \leftarrow K$ users with the largest weighted rate $\lambda_i(t) r_i(t)$;
3: Antenna selection:
successive elimination method as in Algorithm 1 to obtain the set of selected antennas $\mathcal{A}$;
4: Weight updating:
$t = t + 1$;
Update $\lambda_i(t)$ for next time slot according to (7) and (8);
5: Iteration:
repeat steps 2–4 until $t > T_c$;
Output:
the set of selected antennas $\mathcal{A}$ of each time slot;
the set of scheduled users $\mathcal{S}$ of each time slot

It is worth noting that this paper considers a static channel scenario, where the channel is assumed to remain constant throughout the entire communication process. It is also assumed that the base station has already obtained full knowledge of all users' channels through channel estimation, and utilizes the estimated channel information to perform antenna selection and user scheduling. In practical scenarios, however, communication channels are dynamically time-varying due to factors such as user mobility and environmental changes. As user mobility increases, the time-variability of the channel becomes more pronounced, leading to a shorter channel coherence time. In high-mobility environments, the quasi-static channel model no longer holds, resulting in increased channel estimation errors. Moreover, to cope with the rapid channel variation, more frequent channel estimations are required, which consume additional time-frequency resources due to the increased training overhead.

It should be emphasized that the proposed JAS-OUS scheme remains applicable under such conditions, as it assumes that channel estimation has already been performed and that the base station has access to the necessary CSI, which forms the basis for communication. Based on this channel information, antenna and user optimization can be conducted. The estimation process itself is not the focus of this work; instead, we consider only the data

transmission phase. Therefore, as long as the CSI is available, the proposed scheme can be executed.

Of course, to ensure system reliability in practical deployments, it is essential to maintain accurate channel estimation throughout the communication process. Techniques such as shortened CSI update intervals and predictive channel estimation may be employed to track the channel in real time.

4. Simulation Results and Analysis

We assess the effectiveness of the proposed JAS-OUS algorithm and its improved scheme JAS-PFUS, and then provide an evaluation of computational complexity. The channel is modeled as a Rayleigh fading channel. Assuming that a transmitter equipped with 16 antennas is communicating with 16 single antenna users. There are four RF chains in the transmitter, and four target users are selected. Unless otherwise specified, the system parameters are configured as above. All simulations are conducted in MATLAB (R2024b, the MathWorks, Inc., Natick, MA, USA, 2024). The parameters used in the simulations are listed in Table 2.

Table 2. Simulation parameters.

Parameters	Values
Element-wise distribution of channel $\mathbf{H}$	$\mathbf{H}_{ij} \sim \mathcal{CN}(0,1)$
Channel noise $\mathbf{n}$	$\mathbf{n} \sim \mathcal{CN}(\mathbf{0},\mathbf{I})$
Number of base station antennas N_t	16
Number of antennas per user	1
Number of RF chains N_{RF}	4
Total number of users U	16
Number of users to be selected K	4
Scale factor in imperfect CSI δ	0.95 or 0.90
Element-wise distribution of the noise term $\mathbf{\Psi}$ in imperfect CSI	$\mathbf{\Psi}_{ij} \sim \mathcal{CN}(0,\ 1/10)$
Number of time slots for JAS-PFUS T_c	20

4.1. Simulation Results

First, several different non-proportional fairness user scheduling algorithms are compared. The proposed JAS-OUS is compared with joint antenna selection and channel power-based user scheduling (JAS-CPUS) [18] and joint antenna selection and semi-orthogonal user scheduling (JAS-SUS) [15]. JAS-SUS is chosen for comparison because of its superior communication efficiency, while JAS-CPUS has the advantage of extremely reduced computational complexity. Figure 2 illustrates the system sum rate achieved by various algorithms with different SNR. Four antennas are selected from sixteen transmitting antennas while four users are scheduled from sixteen users for message transmission. As shown in the figure, the overall rate of JAS-SUS approximates the sum rate of optimal exhaustive search. The proposed JAS-OUS outperforms JAS-CPUS and is close to the performance of JAS-SUS. For ease of calculation and presentation, the sum rate of JAS-SUS is used as the benchmark for comparison in the following simulations.

In Figure 3, the achievable system throughput of three user scheduling algorithms with various transmitting antenna configurations is compared. The transmitting antennas range from 8 to 16, with the SNR fixed at 10 dB. From the figure, it is evident that as the number of transmitting antennas grows, the system sum rate shows a slow rising trend. When the transmitting antennas increases to a certain number, the sum rate approaches stability.

Figure 4 presents the comparison with respect to the system sum rate using various selected antenna configurations. The quantity of selected antennas is matched to the quantity of allocated users, ranging from 4 to 10, with the SNR fixed at 10 dB. The system sum rate shows a growing trend with the growth in the number of selected antennas, i.e., the more users are served, the greater sum rate will be. The introduced algorithm outperforms JAS-CPUS.

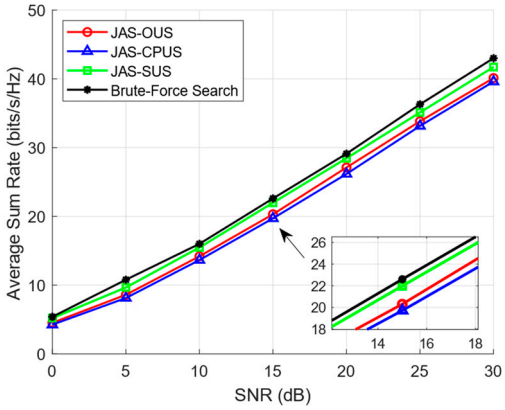


Figure 2. Comparison of the average sum rate at different SNR.

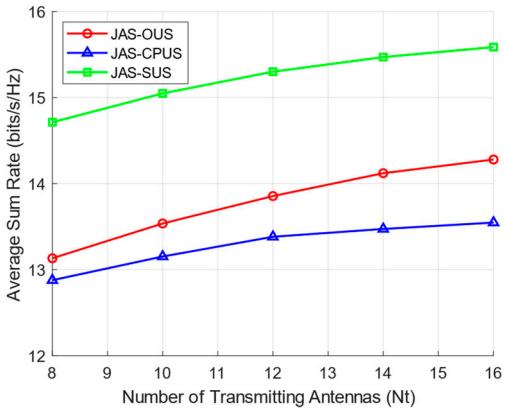


Figure 3. Comparison of the average sum rate with various transmitting antenna configurations.

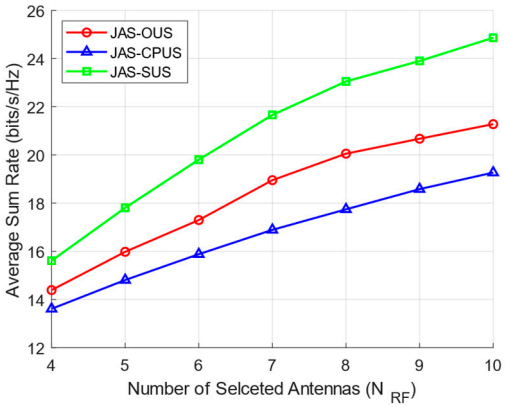


Figure 4. Comparison of the average sum rate with various selected antenna configurations.

Figure 5 illustrates the system throughput with various mobile user configurations. The total number of users is set to range from 8 to 16. Four users are selected from the total for communication, and the SNR is fixed at 10 dB. As the number of users increases, the system sum rate exhibits a slow growth trend. Additionally, the performance gap between JAS-OUS and JAS-CPUS gradually widens, although the overall difference remains relatively small. The performance of JAS-OUS approaches saturation when the

number of users reaches 14, while the performance of the JAS-CPUS algorithm becomes even less sensitive to user number variations. These results indicate that the proposed JAS-OUS algorithm remains applicable even as the number of users increases. However, the system sum rate is not highly sensitive to the number of users, and the gain brought by additional users is marginal. This is because, when only four users are to be selected for communication, even a total of eight users is sufficient to ensure the selection of four users with low channel correlation, thereby achieving a reasonably high sum rate. As the total number of users further increases, although the chance of selecting users with lower correlation improves, it does not result in a qualitative improvement in system performance. Moreover, as the total number of users increases, the computational complexity rises rapidly. Therefore, when only four users need to be selected, a total of eight users in the system strikes a good balance between communication efficiency and computational cost.

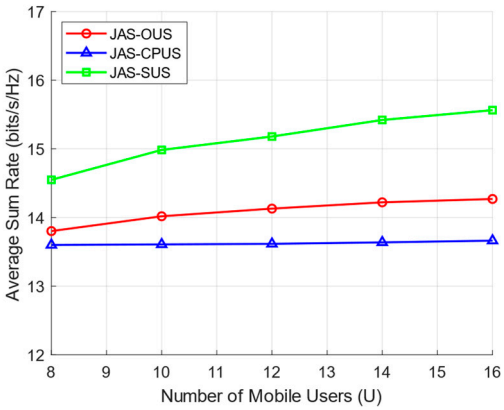


Figure 5. Comparison of the average sum rate with various mobile user configurations.

The above simulations of the JAS-OUS algorithm are based on the assumption of perfect CSI, which is an ideal condition that does not hold in practical scenarios. In reality, channel estimation errors arise due to factors such as environmental variations and user mobility. Therefore, it is necessary to evaluate the performance of the JAS-OUS algorithm under imperfect CSI conditions.

Here, the imperfect estimated channel is modeled as [28]

$$\tilde{\mathbf{H}} = \delta \mathbf{H} + \sqrt{1-\delta^2}\mathbf{\Psi}, \tag{10}$$

where $\mathbf{\Psi}$ is a complex Gaussian noise matrix, whose elements follow the distribution $\mathcal{CN}(0,\ 1/10)$. The parameter $\delta \in [0,1]$ is a scaling factor that characterizes the accuracy of channel estimation. A smaller δ indicates less accurate CSI. When $\delta = 1$, the base station is assumed to have perfect CSI.

As shown in Figure 6, when the SNR is low, imperfect CSI has a limited impact on the system sum rate, which remains close to the value under perfect CSI. However, as the SNR increases, the impact of imperfect CSI becomes more pronounced. For example, under the condition of SNR $= 10$ dB, when $\delta = 0.95$ and $\delta = 0.90$, the system sum rate reaches 97.2% and 93.1%, respectively, of the value achieved with perfect CSI. In contrast, under SNR $= 30$ dB, when $\delta = 0.95$ and $\delta = 0.90$, the system sum rate drops to only 71.9% and 62.7%, respectively, of that under perfect CSI. This is because under imperfect CSI, inter-user interference must be taken into account. At low SNR, the performance is dominated by channel noise, and the degradation due to interference is negligible. However, at high SNR, the impact of noise diminishes, and interference becomes the dominant limiting factor, pushing the system into an interference-limited regime. As a result, under high SNR, the

negative impact of imperfect CSI on the overall signal to interference plus noise ratio (SINR) becomes more substantial, leading to rate saturation or even degradation.

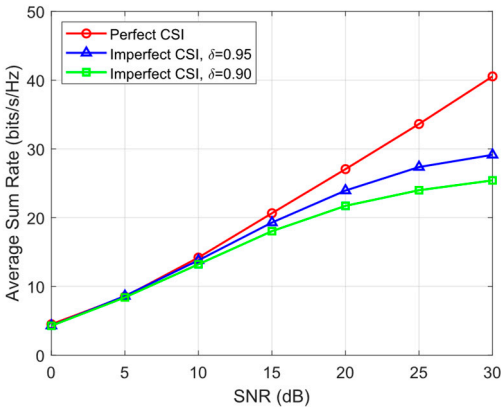


Figure 6. System sum rate under imperfect CSI.

Next, the JAS-PFUS algorithm for fairness scheduling purpose is simulated. In Figure 7, we compare the system throughput of the JAS-PFUS scheme for three different time slots. Both the amount of transmitting antennas and mobile users are 16, with 4 being the amount of target-selected transmitting antennas and users. It can be observed from the figure that, as the SNR increases, the sum rate of the JAS-PFUS algorithm also increases. Moreover, the impact of the number of time slots T_c on the system sum rate is negligible. This is because the window is simply a time-scale parameter that controls the statistical averaging of the transmission rate. As long as each user's average rate reflects its long-term trend, the statistical characteristics of the system scheduling remain stable, and thus the system weighted sum rate shows little variation. However, if the number of time slots is too large, the scheduling response becomes sluggish, leading to situations where some users are not scheduled for extended periods. Conversely, if the number of time slots is too small, the system becomes biased toward users with better instantaneous channel quality, and the fairness of scheduling cannot be maintained. Therefore, in the following simulations, we adopt a compromise value of 20 time slots.

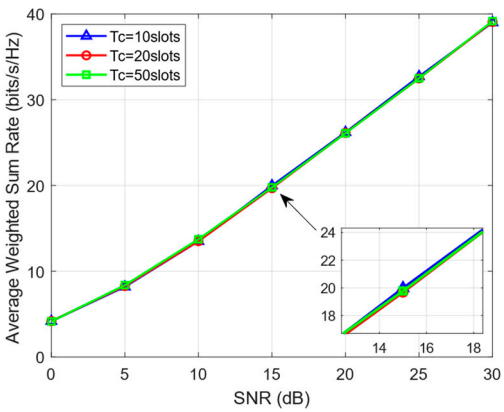


Figure 7. Average sum rate of JAS-PFUS for various time slots.

Figure 8 illustrates the comparison of average weighted sum rate between the non-proportional fairness scheduling algorithm JAS-CPUS and the proportional fairness algorithm JAS-PFUS. JAS-CPUS is an effective algorithm, in which the antenna selection part adopts a successive elimination method, while the user scheduling part is based on the channel power of each user. The weights of all the users are fixed at 1 to calculate the weighted sum rate. In order to fully consider the fairness of users' participation in communication, PFUS algorithm gives each user an adjustable weight and schedules different

users to participate in communication in each time slot by adaptively adjusting the weights. As can be seen from the figure, JAS-PFUS only slightly reduces the weighted sum rate. The efficiency loss is very small, which ensures the effectiveness of the transmission.

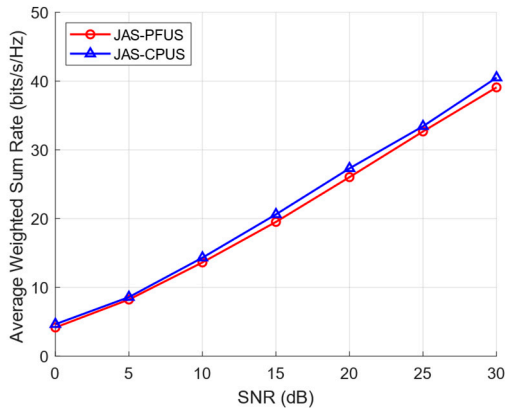


Figure 8. Average weighted sum rates comparison between JAS-PFUS and JAS-CPUS.

In Figure 9, we compare the fairness of two algorithms in different scenarios, where the fairness metric is average rate per user. The simulation setup is 20 time slots, 16 total users, and 10 dB SNR. We compare the fairness of two algorithms by calculating the average rate of each user over 20 time slots. A user has a positive rate only when he is scheduled. As illustrated in Figure 9a, JAS-PFUS algorithm is capable of scheduling 15 users to participate in communication within a time duration of 20 time slots, while the algorithm without proportional fairness can only schedule 4 users. In Figure 9b, the number of RF chains increased to six. Assuming that the quantity of selected users equals , i.e., the quantity of users allocated is also six. JAS-PFUS can be still capable of scheduling up to 14 users. In Figure 9c, by limiting the total number of users to 10, JAUS-PFUS is able to schedule all users to participate in the communication. Thus, regardless of whether the quantity of selected participants is increased or the total quantity of participants is decreased, the proposed algorithm is able to schedule more users, which fully realizes fairness in user scheduling.

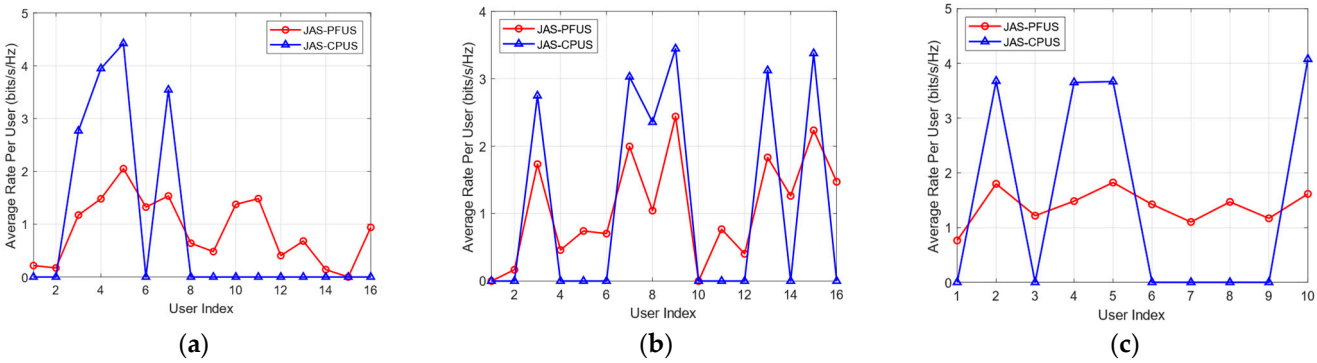


Figure 9. Comparison of Fairness between JAS-PFUS and JAS-CPUS: (**a**) $SNR = 10$ dB, $N_t = U = 16$, $N_{RF} = 4$; (**b**) $SNR = 10$ dB, $N_t = U = 16$, $N_{RF} = 6$. (**c**) $SNR = 10$ dB, $N_t = 16$, $U = 10$, $N_{RF} = 4$.

4.2. Computational Complexity Analysis

We compare the proposed JAS-OUS algorithm with JAS-CPUS, JAS-SUS, and the exhaustive search method. The algorithms JAS-OUS, JAS-CPUS, and JAS-SUS belong to the same class of methods, where the antenna selection part is implemented using the successive elimination method, and the difference in complexity lies only in the user scheduling algorithm employed.

For the JAS-OUS algorithm, a total of $N_t - N_{RF}$ iterations are required, where in each iteration one antenna is removed until only N_{RF} antennas remain. As shown in Algorithm 1, the main computational burden lies in step 3. In the ith iteration, $i-1$ antennas have already been removed, so the channel matrix $\tilde{\mathbf{H}}$ in step 3 has a dimension of $U \times (N_t - i)$. The OUS algorithm (Algorithm 2) is applied to $\tilde{\mathbf{H}}$, where in each iteration, one user is added to the scheduling set. The complexity of Algorithm 2 is $\mathcal{O}(\sum_{n=1}^{K-1}(U-n)(N_t-i)n)$. Subsequently, step 3 in Algorithm 1 computes the system sum rate $R_{sum}(\mathcal{A}\backslash\{i\},\mathcal{S})$ for the selected users. This involves ZF precoding and water-filling power allocation, where the complexity of ZF precoding is $\mathcal{O}(K^2(N_t-i)+K^3)$, and the complexity of water-filling is $\mathcal{O}(K\log K)$. Therefore, the total complexity of the JAS-OUS algorithm is $\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}(\sum_{n=1}^{K-1}(U-n)(N_t-i)n+K^2(N_t-i)+K^3+K\log K))$. After removing lower-order terms and constant factors, the simplified complexity is $\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}(UK^2-K^3)(N_t-i))$. Similarly, the computational complexity of JAS-CPUS is $\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}((U+K^2)(N_t-i)+K^3))$, and that of JAS-SUS is $\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}(UK^2-K^3)(N_t-i))$. In the exhaustive search method, all possible combinations of N_{RF} antennas from N_t candidates and K users from U total users are enumerated. There is a total of $C_{N_t}^{N_{RF}}C_U^K$ combinations. For each combination, ZF precoding and water-filling power allocation are performed to compute the system sum rate. Therefore, the overall complexity of the exhaustive search is $C_{N_t}^{N_{RF}}C_U^K(K^2N_{RF}+K^3)$.

It can be observed that JAS-OUS and JAS-SUS have similar computational complexity, while JAS-CPUS has a relatively lower complexity. This is because CPUS directly compares the channel norms of all users and selects the top K users with the largest norms, without any iterative process. In contrast, JAS-OUS iteratively selects users by evaluating the channel orthogonality between each remaining user and the set of already selected users, choosing the one with the highest orthogonality at each step. This process iterates $K-1$ times, which contributes to the higher complexity. The exhaustive search clearly has the highest complexity but guarantees the optimal solution. The computational complexities of different algorithms are summarized in Table 3.

Table 3. Computational complexity comparison.

Algorithms	Computational Complexity
JAS-OUS	$\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}(UK^2-K^3)(N_t-i))$
JAS-CPUS	$\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}((U+K^2)(N_t-i)+K^3))$
JAS-SUS	$\mathcal{O}(\sum_{i=1}^{N_t-N_{RF}}(UK^2-K^3)(N_t-i))$
Exhaustive Search	$C_{N_t}^{N_{RF}}C_U^K(K^2N_{RF}+K^3)$

In addition, we compare the execution time required by the above algorithms under a single channel realization. Since the simulation time is influenced by both the software version and computer hardware configuration, we provide the corresponding details in Table 4. All simulations are conducted on a personal computer using MATLAB R2024b. Table 5 compares the simulation time required by different user scheduling algorithms in a single channel realization. It can be seen that JAS-CPUS has the smallest time overhead.

Although the proposed JAS-OUS algorithm takes more simulation time than the JAS-CPUS algorithm, it achieves better system sum rate. Therefore, JAS-OUS achieves a compromise between computational complexity and communication efficiency. Further, the proposed JAS-PFUS performs user scheduling in all time slots to let more users participate in communication at the cost of more simulation time.

Table 4. Simulation software and hardware configuration.

Software and Hardware	Configuration Parameters
Operating System	Windows 10 Professional
Simulation Software	R2024b
CPU	Intel(R) Core(TM) i9–9880H
RAM	16 GB
GPU	Radeon Pro 560X
VRAM	4096 MB

Table 5. Simulation time for one channel realization.

Algorithms	Simulation Time
JAS-OUS	0.1640
JAS-CPUS	0.1029
JAS-SUS	0.2087
JAS-PFUS	1.4069
Exhaustive Search	1808.4610

5. Conclusions

In this work, we study the joint antenna selection and user allocation problem for multi-user large-scale MIMO systems. In massive MIMO systems, antenna selection and user scheduling are complementary techniques. Antenna selection reduces hardware complexity and energy consumption, while user scheduling enhances system capacity and fairness. Together, they form the core mechanism for system performance optimization, serving as key enablers for the successful deployment of massive MIMO.

In this paper, a low-complexity joint antenna selection and user scheduling scheme is designed for a multi-user system model, which achieves a considerable system sum rate. On this basis, to balance system performance and user service fairness, an improved algorithm based on PF scheduling is proposed. This method ensures fair participation of users in communication with only a slight loss in transmission rate, making it practically meaningful.

The integration of antenna selection and user scheduling serves as a key framework for resource allocation algorithms in wireless communication systems. It enhances the system's resilience to hardware failures and non-ideal channel conditions, offering a low-complexity, suboptimal alternative to fully connected architectures. This approach is particularly suitable for power-sensitive scenarios such as edge devices, UAVs, and satellite communications, as well as a wide range of multi-user access systems including 5G/6G, IoT, and Wi-Fi networks. Therefore, joint antenna selection and user scheduling holds profound theoretical research value and promising practical significance for future deployments.

Author Contributions: Conceptualization, M.W.; methodology, B.C.; software, B.C. and M.W.; validation, L.Y. and M.W.; formal analysis, L.Y.; investigation, B.C.; resources, L.Y.; data curation, L.Y.; writing—original draft preparation, B.C., L.Y. and M.W.; writing—review and editing, B.C., L.Y. and M.W.; visualization, B.C. and M.W.; supervision, M.W.; project administration, M.W.; funding acquisition, L.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Natural Science Foundation, grant number 62372244, 62172235; the National Key Research and Development Program of China, grant number 2021YFB3101100; the Primary Research & Development Plan of Jiangsu Province, grant number BE2023025; the Natural Science Foundation of Nanjing University of Posts and Telecommunications, grant number NY222132; the ZTE Industry-university Research Fund, grant number 2023ZTE08-02; the Postgraduate Research & Practice Innovation Program of Jiangsu Province, grant number 46004CX17781, SJCX220285.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

Acronyms	Full Forms
MIMO	Multi-Input Multi-Output
RF	Radio Frequency
JAS-OUS	Joint Antenna Selection and Orthogonality-based User Scheduling
PFUS	Proportional Fairness User Scheduling
IoT	Internet of Things
V2X	Vehicle-to-Everything
SNR	Signal-to-Noise Ratio
UAV	Unmanned Aerial Vehicle
MGEM	Matrix Gaussian Elimination Method
CSI	Channel State Information
POMDP	Partially Observable Markov Decision Processes
AMCMC	Adaptive Markov Chain Monte Carlo
NOMA	Non-Orthogonal Multiple Access
OFDMA	Orthogonal Frequency Division Multiple Access
JASUS	Joint Antenna Selection and User Scheduling
QoS	Quality of Service
JAS-CPUS	Joint Antenna Selection and Channel Power based User Scheduling
JAS-SUS	Joint Antenna Selection and Semi-orthogonal User Scheduling

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